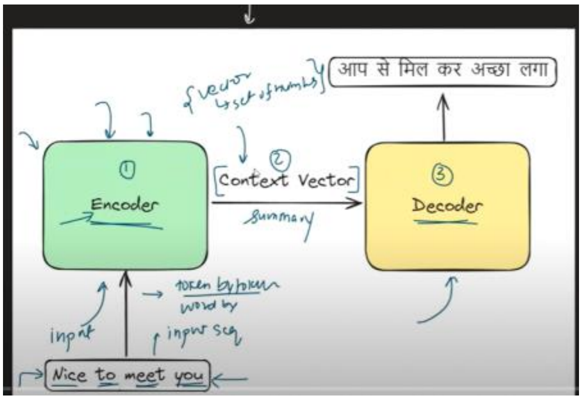
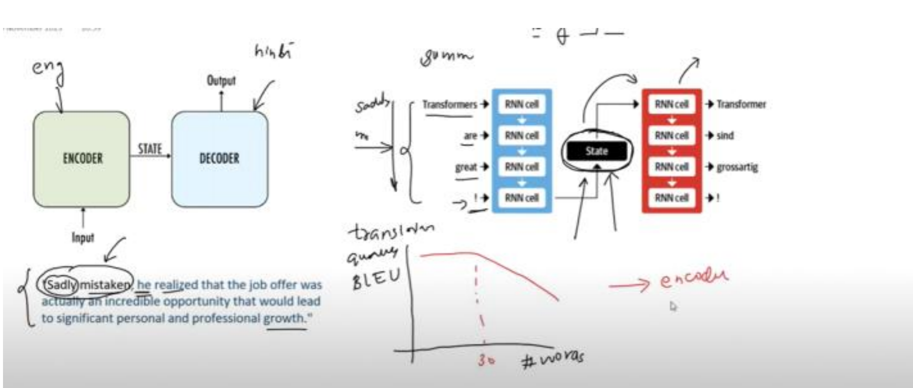
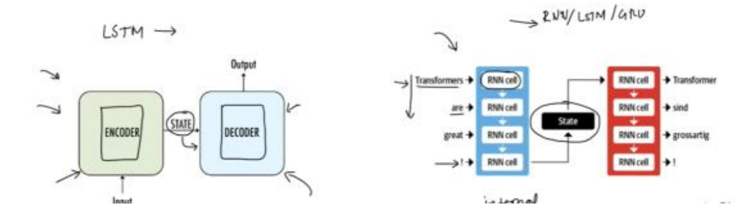
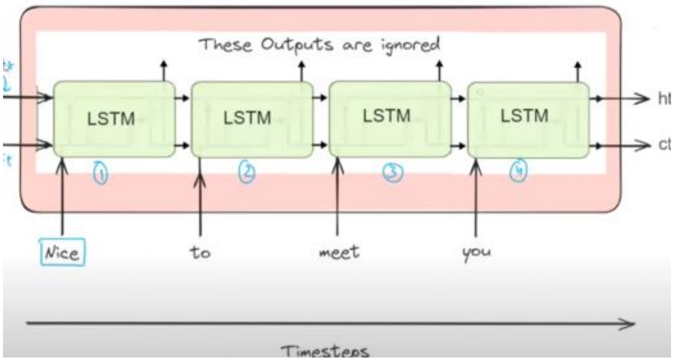
Encoder & Decoder

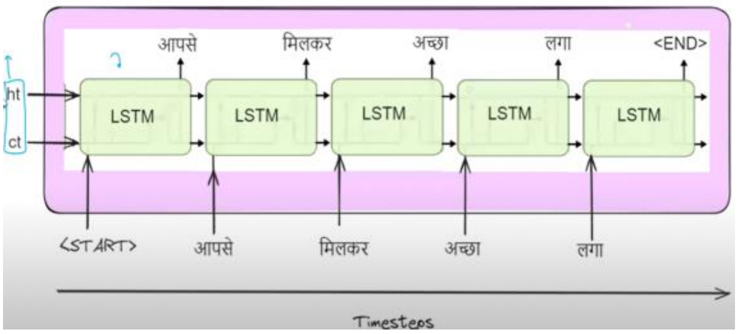
30 May 2024 07:33



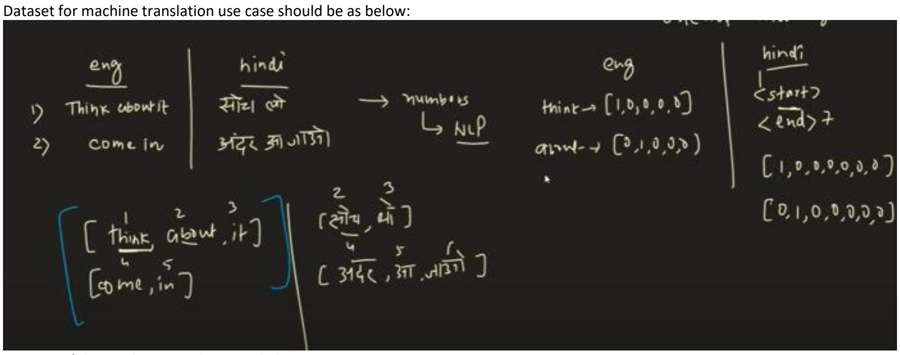




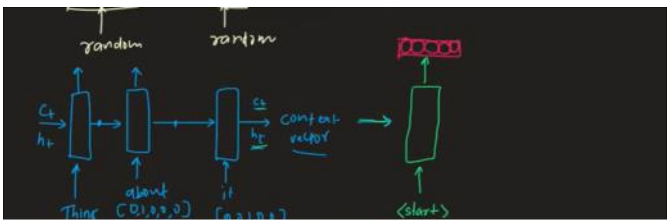


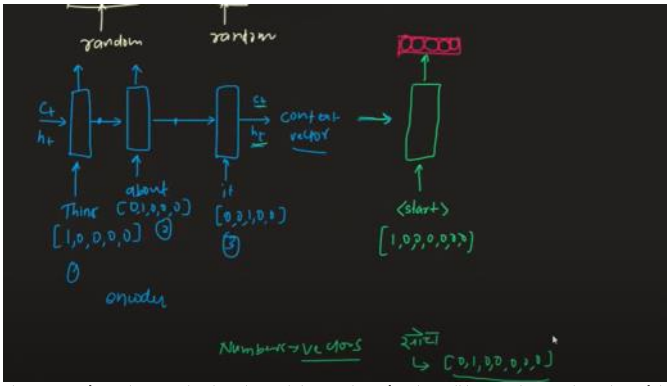


Dataset for machine translation use case should be as below:

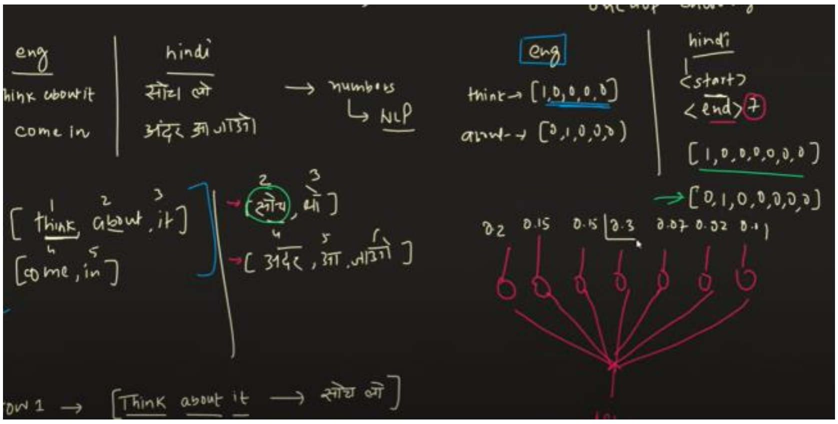


Training of the machine translation as below:

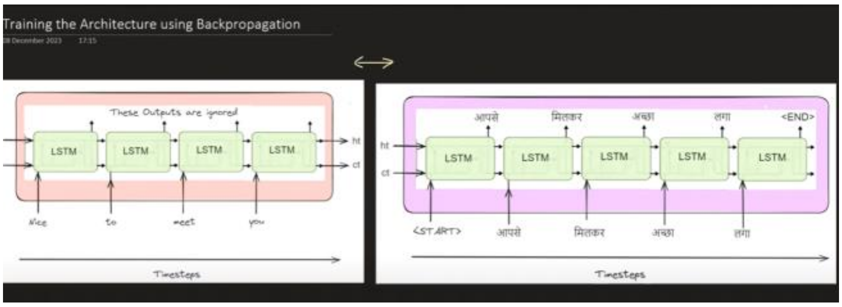




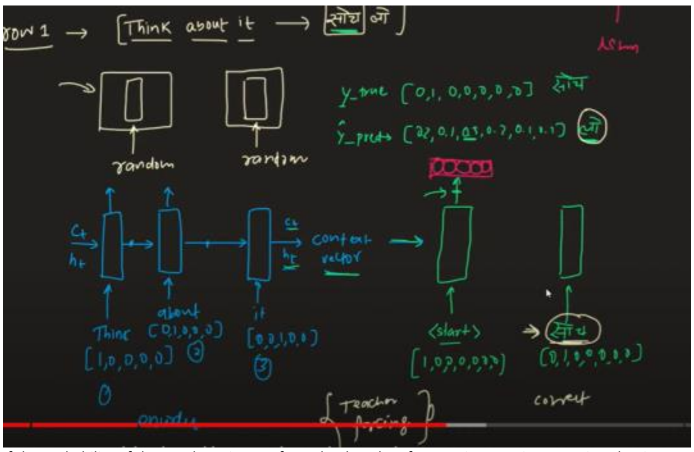
There is a softmax layer in the decoder and the number of nodes will be equal to total number of the translated words



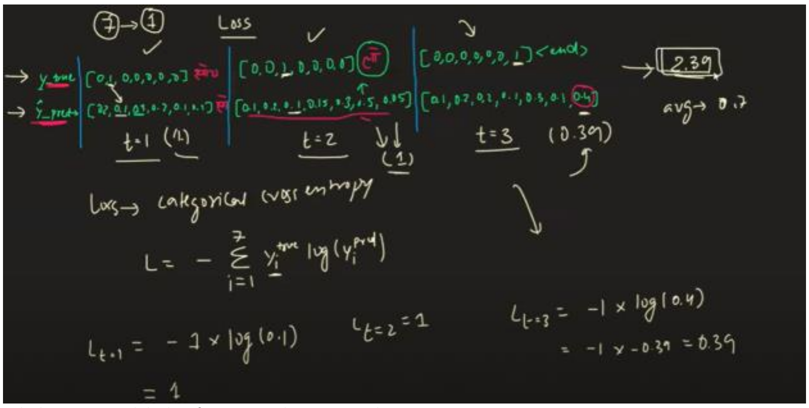
Encoder



Decoder



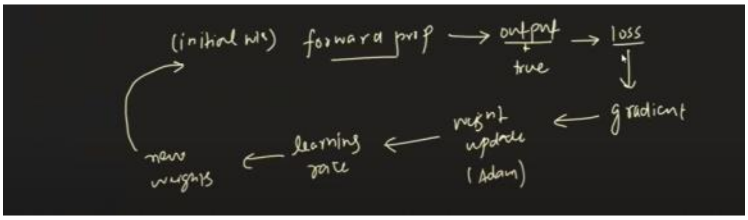
If the probability of the word coming out from the decoder after 1st timestep is wrong, i.e. yhat is not equal to y, in that case also basis the translated dataset we will send the correct translated word as input to the decoder in the next timestep . This technique is called **teacher forcing**. Because of this the training will be faster.

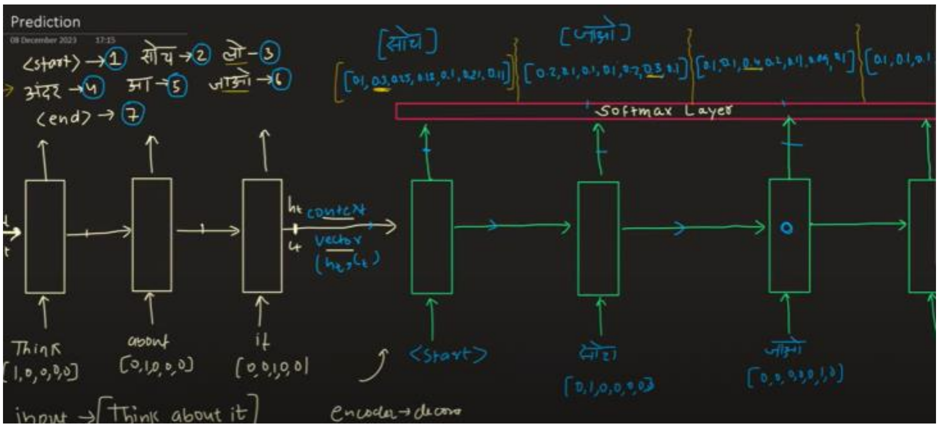


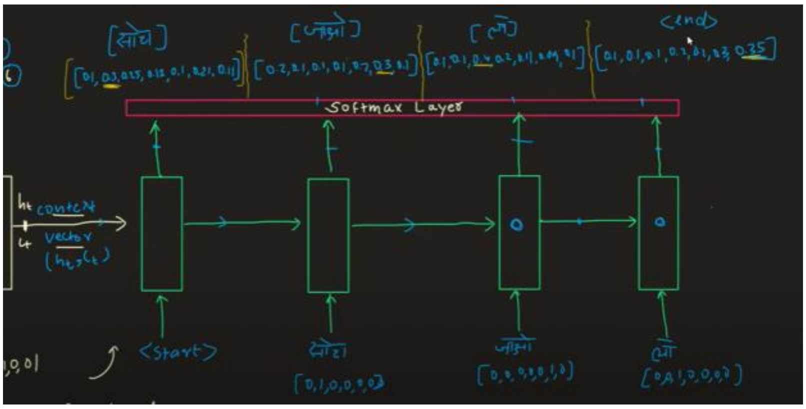
Calculating Loss and the loss function used is Categorical cross entropy

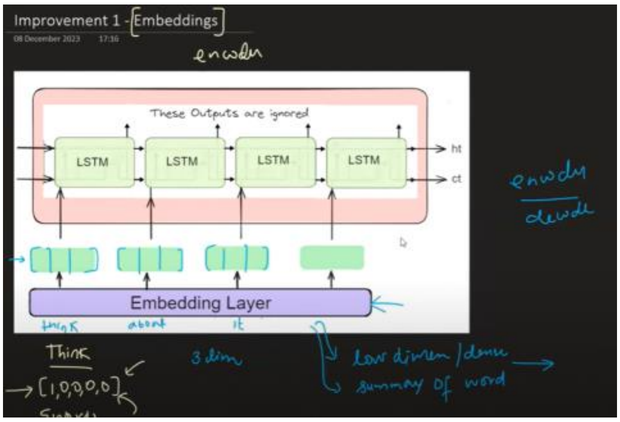
Back Propagation of Encoder and Decoder as below:

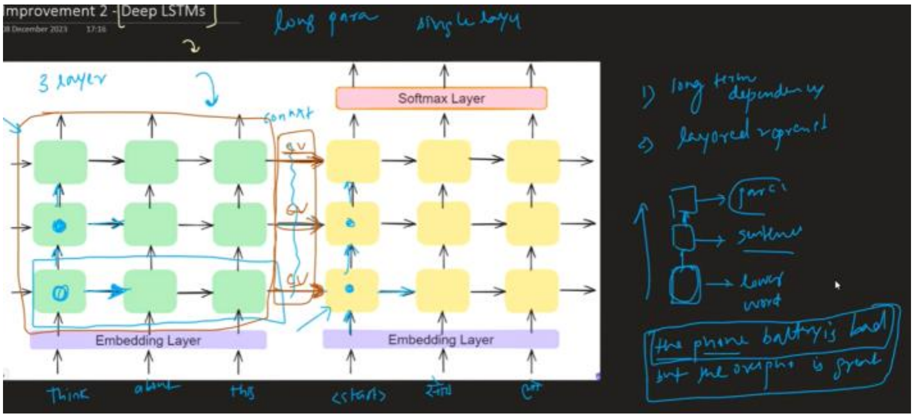
Gradient - helps to find the contribution of particular word vector towards the loss. Then using the gradient, weights of the parameters are update. The update is done by the Optimizers



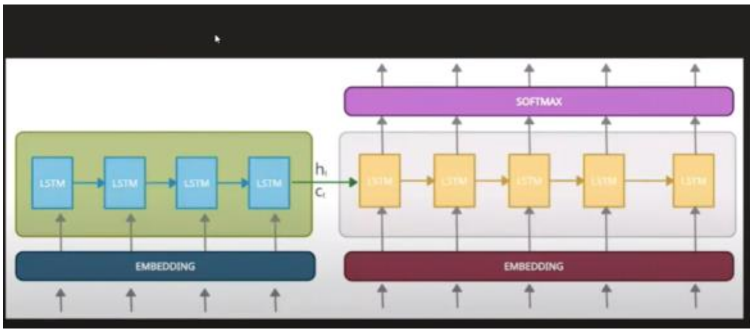


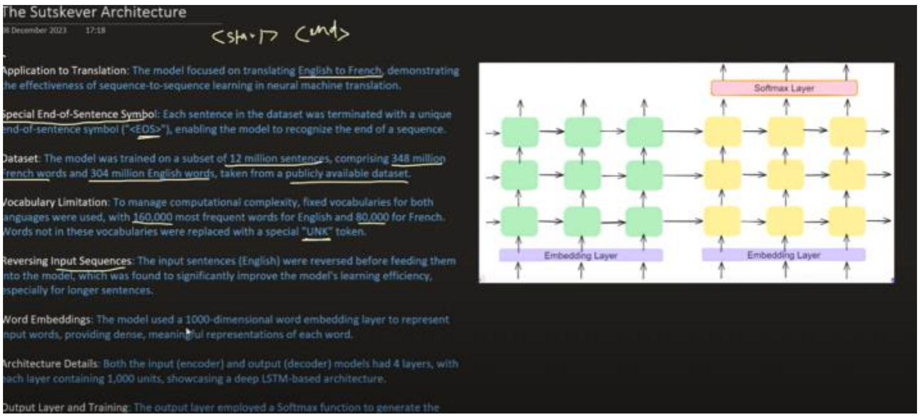






Multiple layers of LSTM in encoder and decoder should be used in case large paragraphs. It works better for long paragraphs, also understands the hierarchy of the data (words -> sentence --> paragraph)





end-of-sentence symbol ("<EOS>"), enabling the model to recognize the end of a sequence.

Dataset: The model was trained on a subset of 12 million sentences, comprising 348 million French words and 304 million English words, taken from a publicly available dataset.

Vocabulary Limitation: To manage computational complexity, fixed vocabularies for both languages were used, with 160,000 most frequent words for English and 80,000 for French. Words not in these vocabularies were replaced with a special "UNK" token.

Reversing Input Sequences: The input sentences (English) were reversed before feeding them into the model, which was found to significantly improve the model's learning efficiency, especially for longer sentences.

Word Embeddings: The model used a 1000-dimensional word embedding layer to represent input words, providing dense, meaningful representations of each word.

Architecture Details: Both the input (encoder) and output (decoder) models had 4 layers, with each layer containing 1,000 units, showcasing a deep LSTM-based architecture.

Output Layer and Training: The output layer employed a Softmax function to generate the probability distribution over the target vocabulary. The model was trained end-to-end with these settings.

Performance - BLEU Score: The model achieved a BLEU score of 34.81, surpassing the baseline Statistical Machine Translation (SMT) system's score of 33.30 on the same dataset, marking a significant advancement in neural machine translation.